Check for updates

ASME Journal of Engineering for Gas Turbines and Power
Online journal at:
https://asmedigitalcollection.asme.org/gasturbinespower

Characterization of a Fluidically Variable Swirl Burner: Effects of the Imprinted Swirl on Flame Shapes and Emissions for Hydrogen–Methane Blends

Mattias E. G. Eck[1]
Chair of Fluid Dynamics,
Technische Universität Berlin,
Berlin 10623, Germany
e-mail: m.eck@tu-berlin.de

Philipp zur Nedden
Chair of Fluid Dynamics,
Technische Universität Berlin,
Berlin 10623 Germany

Jakob G. R. von Saldern
Laboratory for Flow Instabilities and Dynamics,
Technische Universität Berlin,
Berlin 10623 Germany

Alessandro Orchini
Chair of Nonlinear Thermo-Fluid Mechanics,
Technische Universität Berlin,
Berlin 10623 Germany

Christian Oliver Paschereit
Chair of Fluid Dynamics,
Technische Universität Berlin,
Berlin 10623 Germany

Gas turbine combustors rely on swirling flows for flame stabilization. The flow's swirl number is herein characterized by the burner's swirler geometry. In addition to stability, swirl represents an important parameter affecting emissions, flame shapes, and flame dynamics. The swirl dependency is furthermore gaining importance with the progression of hydrogen-enriched fuels. Hydrogen flames are well stabilized by axial jets, whereas a stable methane combustion is optimally maintained by swirled flows. The active variation of swirl therefore plays a significant role and represents a cornerstone of this study. Here, the experimental characterization of a novel swirl-stabilized burner is presented. It allows for the variation of the swirl number through fluidic actuation, not employing geometry variations, and capable of creating a linearly controllable flow regime ranging from a nonswirled to a fully swirled flow. This concept is characterized under reacting conditions. For a broad range of hydrogen–methane gas mixtures, the characteristics of the burner and the stabilized flames are experimentally investigated. For each fuel blend, a wide span of operating points is created with different swirl regimes. The exhaust emissions are analyzed through a gas analyzing system, and the respective flame shapes are acquired through OH-imaging. The characterization of the novel burner concept demonstrates its operational capability over a broad operational range. For each fuel composition, an optimal degree of swirl could be identified, allowing for a stable combustion, and minimizing the emissions. Stable conditions are mapped, quantifying the operational limits of the investigated burner.*
[DOI: 10.1115/1.4069726]

Keywords: combustor design, fluidic actuation, hydrogen combustion

1 Introduction

The operation of modern gas turbines strictly depends on the stability of the respective combustion process. To achieve flame stabilization, gas turbine combustors typically rely on swirling flows, which are generated through suitable swirlers, generally integrated in the burner geometry [1–3]. The effects of swirling flows on the combustion process are significant and, thus, are still focused upon by extensive research throughout academia and industry. In addition to mere flame stabilization and anchoring, the degree of imprinted swirl affects, among others, the air-fuel mixing efficiency, heat transfer mechanisms, and, especially, the static and dynamic stability of flames [1,3–7]. The multifaceted effect of swirl deserves particular interest since it is of great importance in the contemporary designation of decarbonizing the energy sector and, consequently, for the combustion of highly hydrogen-enriched fuel blends. For a stable and clean combustion, the degree of swirl is chosen based on the combusted fuel blend. Generally, the swirl is decreased with increasing hydrogen content, possibly combined with the use of pilot flames or steam and inert gas dilution [8,9].

The ability to vary the incited swirl, quantified by the dimensionless swirl number, provides a valuable research contribution, particularly for studying flame dynamics, thermoacoustic instabilities, and their underlying effects [3]. Multiple approaches are known which allow for the variation of the swirl number of swirl-stabilized combustors, generally relying on physical flow interference, mechanical actuation, or geometrical variation of the respective swirler segments (e.g., see Refs. [3,10–12]). Correspondingly, the burner concept characterized in this work makes use of fluidic actuation, allowing for an active and continuous swirl number variation. Its operation solely relies on the controlled injection of a secondary air mass flow, thus allowing for the abrogation of any movable parts or geometry variations. The underlying working principle exploits the Coanda effect, and has been adopted from the field of aircraft propulsion systems where it is

Turbo Expo: Turbomachinery Technical Conference & Exposition (GT2025) June 16–20, 2025. GT2025.
[1]Corresponding author.
Manuscript received June 30, 2025; final manuscript received July 21, 2025; published online December 4, 2025. Editor: Jerzy T. Sawicki.

Copyright © 2026 by ASME; reuse license CC-BY 4.0

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/1/011024/7536570/gtp-25-1257.pdf by guest on 01 January 2026

studied as incorporation within thrust vectoring systems (e.g., see Refs. [13–16]). The burner concept and its geometry are discussed in more detail in Sec. 2.1.

The burner design, its working principle, and, particularly, its operational behavior have been assessed and investigated in previous studies. In this regard, its capability of varying the degree of swirl solely relying on fluidic actuation has been qualitatively ascertained in a numerical study [17], and subsequently demonstrated and quantified experimentally under cold, i.e., nonreacting conditions, see Ref. [18]. Among others, a governing factor has been derived, controlling the magnitude of the imprinted swirl: the ratio between the injected secondary and the primary air mass flow, defined as the mass flow ratio (MFR). Figure 1 shows the dependency of the dimensionless swirl number S, acquired experimentally in nonreacting conditions, on the set MFR. This relation represents a summary of the cold characterization of the concept, more details of which are published in Ref. [18].

For low MFR values (MFR $< 5\%$), the induced swirl is subjected to strong fluctuations and, depending on the operating conditions, is subjected to hysteretic behavior. However, at $\mathrm{MFR}_{0,\mathrm{cold}} = 6\%$, a stable and reproducible condition is identified, at which no swirl is imparted, resulting in a swirl number of 0 (in the following, MFR values leading to a zero-swirl regime are denoted as MFR_0). With further increasing MFR, a region with near-linear control over S is reached, terminating at MFR $\approx 32\%$, where no further increase of the swirl magnitude is achieved by further increasing the MFR. The near-linear control region, as well as the stable zero-swirl conditions, are found to yield significant relevance for advancing the development of the novel burner. This allows for the further comprehension of the heterogeneous effects of swirl on combustion and related technical applications.

The objective of this study is to examine whether the achievable swirl number range observed under nonreacting conditions can be effectively exploited to enable fuel flexibility in reacting conditions. To this end, different swirl regimes ranging from a pure axial jet to a fully swirled flow span a wide spectrum of operating conditions. Fuel flexibility is introduced by operating with different fuel mixtures of hydrogen and natural gas (NG). For each operating point, flame shape and stability are assessed through the evaluation of OH* images. For the operational map spanned by swirl and hydrogen–NG mixture, the flame flashback limits are identified. Additionally, the emissions are analyzed within the exhaust gas through a gas analyzing system. Results are discussed and compared to similar industrial burner concepts, highlighting the technical relevance of the presented fluidically variable swirl burner.

2 Experimental Investigation

This section describes the experimental investigation of the fluidically variable swirl burner. The burner model and the combustion test rig used to characterize it are presented, and the measurement techniques and the chosen operating range are illustrated.

2.1 Fluidically Variable Swirl Burner. The novel fluidically variable swirl-stabilized burner characterized in this work allows for

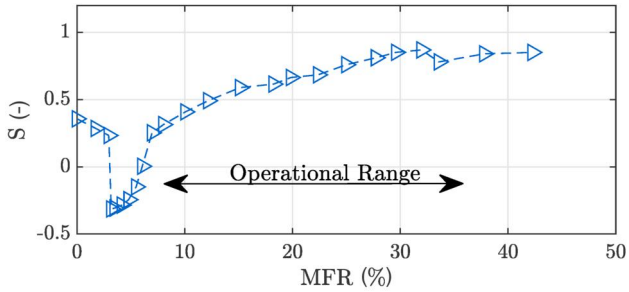

Fig. 1 Original relation between swirl number *S* and MFR (nonreacting, adapted from Ref. [18])

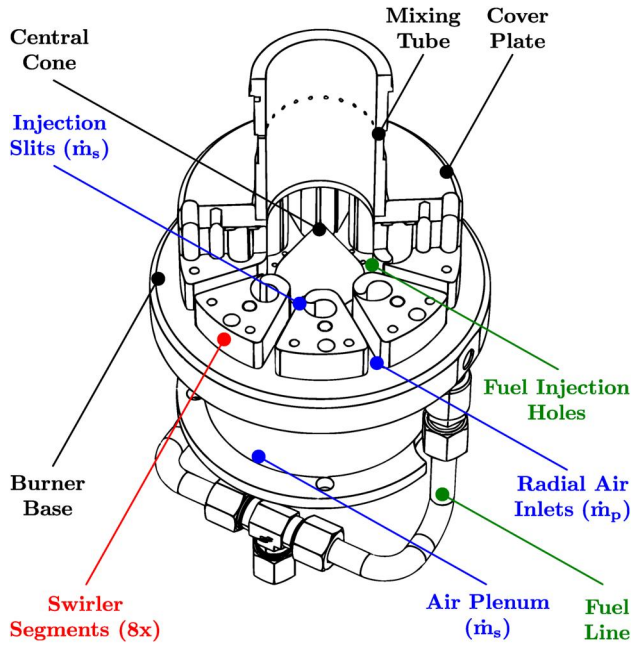

Fig. 2 Section view of the burner prototype. Key-features, as air supply and fuel supply components, are highlighted (flow from bottom to top).

the active and continuous variation of the degree of imprinted swirl. It solely relies on fluidic actuation, and does not comprise any moving parts nor features geometric variations. For the operational characterization under reacting conditions, a functional burner prototype has been developed based on the original concept, as schematically depicted in Fig. 2. It comprises fuel injection ports, located on the base plane of the swirler, a secondary air injection, and eight circumferentially arranged swirler segments, each of which features injection slits. The secondary air injection can be regulated to vary the flux of air through the thin slits, by which a wall jet can be created. The secondary air is fed into the segments through a common plenum, positioned upstream of the burner base. The segments are positioned and mounted between the burner base and a top plate. The primary air is fed into the burner radially, hence flowing through the eight channels between the aforementioned segments. The inflowing primary air is here impacted by the wall jets and is subjected to a directional change. The deflection magnitude and thus the efficiency of the fluidically-variable swirl concept rely on air entrainment and on the leverage on flow separation effects via the Coanda effect. By varying the secondary air mass flowrate ($\dot{m}_s$), while keeping the total mass flowrate of combustion air ($\dot{m}_{tot}$) constant, the MFR, defined as the ratio between primary ($\dot{m}_p$) and secondary mass flow rates, is varied:

$$\mathrm{MFR} \equiv \frac{\dot{m}_s}{\dot{m}_p}, \quad \text{with} \quad \dot{m}_{tot} = \dot{m}_p + \dot{m}_s \tag{1}$$

The MFR is found to be a governing parameter for the induced swirl and represents a key parameter affecting the operation of the fluidically variable swirl burner. Toward the center of the burner's swirler segment, the air is directed toward the mixing tube and the burner's outlet through a central cone, at the base of which the fuel injection is located.

2.2 Test Rig. The combustion test rig used in the experimental investigation is schematically depicted in Fig. 3. It consists of an open circular air duct in which the preheated primary air flow is injected. Through a separate air line, the nonpreheated secondary air mass flow is injected into the burner. The fuel is supplied through a main fuel line. Various fuel gas blends, produced in a fuel gas mixing

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/1/011024/7536570/gtp-25-1257.pdf by guest on 01 January 2026

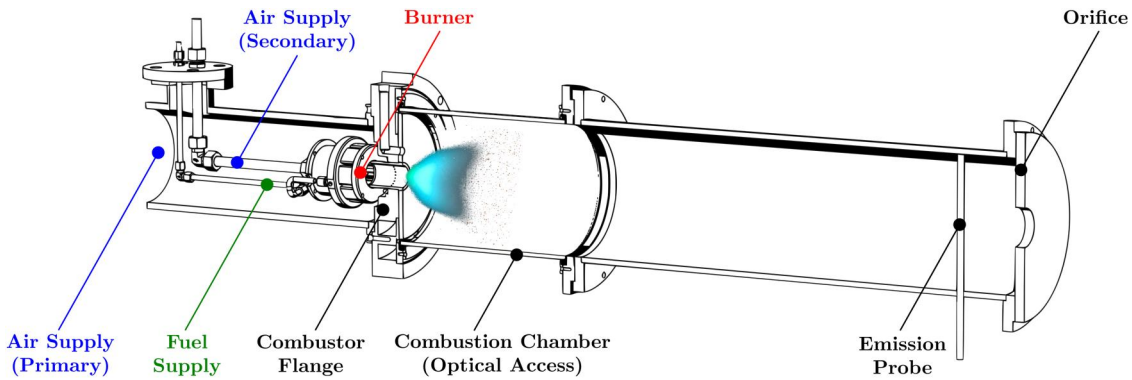

Fig. 3 Schematic overview of the combustion test rig (flow from left to right)

station, can be fed to the system, and the fuel composition can be varied during operation. Downstream the burner's mixing tube outlet, a quartz glass combustion chamber with an inner diameter of 200 mm and a length of 300 mm is mounted, providing optical access to the flame.

2.2.1 Measurement Equipment. In addition to temperature measurements at several positions along the entire test rig, the following diagnostic tools are used to characterize the flame response and combustion properties:

OH*-imaging: For monitoring the flame shapes and their stability, the OH* chemiluminescence images are recorded through an intensified camera,[2] sensitive to the ultraviolet spectrum, and equipped with an optical bandpass filter allowing for the transmission of emitted light around a wavelength of $\lambda = 308$ nm. The camera is setup to yield an image resolution of up to 1200×1900 pixels at a constant frame rate of 161 frames per second. The OH* intensity can be assumed to relate to the local heat release rate of the flame [19,20], thus determining its qualitatively shape independently from the specific fuel blend choice. From the OH* chemiluminescence recording mean flame shapes are obtained by averaging the time series over 5 s (amounting to 805 frames).

Emissions analysis: As depicted in Fig. 3, at the test rig's downstream termination, exhaust gases are probed through a specifically designed emissions probe that allows for the sampling of gaseous probes over the entire exhaust tube diameter. The sampled exhaust gas is led to the analyzing system[3] through a heated hose pipe. Remaining condensate is removed and the sample directed to different analyzing stages, where, on a dry basis, concentrations of NO_x, O_2,[4] CO,[5] and CO_2 (Footnote 5) are determined. NO_x concentrations are herein quantified through chemiluminescence analysis within an apposite section, also distinguishing between NO and NO_2,[6] whereas CO and CO_2 concentrations are assessed through their specific absorption behavior. In order to obtain comparable results, the raw readings of the analyzing systems are normalized (see Sec. 3.2).

2.3 Operating Conditions. The employed atmospheric combustion test facility allows for the generation of a wide range of operating conditions by allowing for variations of the fuel composition and set MFR values. While the total air mass flow participating in the combustion process is kept constant at $\dot{m}_{\text{tot}} = 175$ kg/h, the MFR is varied, thus generating a variety of flow regimes characterized by different degrees of swirl. The used fuel gas is a blend between natural gas (NG, with a methane content of approx. 90% mol.) and hydrogen (H_2), where the H_2 content is quantified by the respective mass fraction X_{H_2}. The primary air flow $\dot{m}_p$ is preheated, while the secondary air mass flow $\dot{m}_s$ remains nonpreheated. In order to realize a constant air temperature of 300 °C at the outlet of the burner's mixing tube, the preheating temperature of $\dot{m}_p$ is increased and adjusted to the respective MFR value to account for the variable mixing ratio between the hot $\dot{m}_p$ and cold $\dot{m}_s$. The burner is characterized by two different thermal power outputs, $P_{\text{th}} = 80$ kW and 100 kW, realized through the variation of the total mass of injected fuel, based on the lower heating value of the used fuel blend. This results in a range of equivalence ratios ϕ between approximately 0.47 and 0.71, as depicted in Fig. 4. The air mass flow $\dot{m}_{\text{tot}}$ is kept constant. The parameter range investigated in this work is summarized in Table 1.

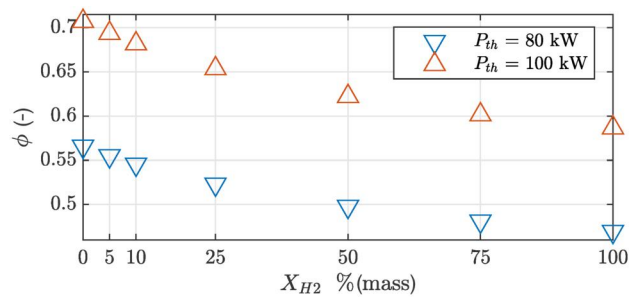

Fig. 4 Range of equivalence ratios

3 Experimental Results

This section summarizes and discusses selected experimental results. We start with presenting the acquired flame shapes based on the OH* chemiluminescence imaging. In the second part, emission levels are investigated as the operating conditions are varied.

3.1 Flame Shapes. Figure 5 exemplary shows flame images of the burner's operation at a thermal power output of $P_{\text{th}} = 100$ kW at a hydrogen content of $X_{H_2} = 50\%$ for two different swirling conditions, MFR = 9% and 16.67%. The top row shows images within the visible spectrum, while the bottom row shows the

Table 1 Varied operational parameters

Parameter	Range
Combustion air	$\dot{m}_{\text{tot}} = 175$ kg/h with $\dot{m}_{\text{p}}$ preheated
Mass flow ratio	$6\% \leq \text{MFR} \leq 36\%$
Thermal power	$P_{\text{th}} = 80$ kW, 100 kW
H_2 content (mass fraction)	$0\% \leq X_{H_2} \leq 100\%$
Equivalence ratio	$0.47 \leq \phi \leq 0.71$

[2]Lambert Tricam, Lambert Instruments B.V, Groningen, The Netherlands.
[3]System by ABB Automation GmbH, Ratingen, Germany.
[4]ABB, *Magnos 106*.
[5]ABB, *Uras 14*.
[6]Knestel CLD dual Basic, KNESTEL Technologie & Elektronik GmbH, Hopferbach, Germany.

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/1/011024/7536570/gtp-25-1257.pdf by guest on 01 January 2026

corresponding time-averaged OH* images. It can be observed that the flame is of conical shape for the low MFR value and changes to a V-shaped flame for the higher MFR value. This demonstrates that the fluidically variable swirl burner can generate a sufficiently large range of swirl numbers to produce distinctly different flame shapes.

For a more detailed analysis, Fig. 6 presents the mean flame shapes for selected operating conditions at different MFR values. The flame shapes are exemplary shown for the operation at 100 kW of thermal power output, for three characteristic hydrogen mass fractions within the combusted fuel blend: $X_{H_2} = 0\%$, 10%, and 100%. A significant variety of different flame shapes can be observed, depending on the set MFR value and hydrogen content. In the following, the effect of individual parameter variations is discussed.

3.1.1 Zero-Swirl Regime. Particularly, the flame shapes acquired for the condition for MFR = 9% is notable, as the combustion appears to take place in an axial jet flame, not comprising any swirl. In this respect, the flame lengths derived from OH* images recorded for MFR = 9% amount to approximately $l_{fl}/D \approx 2$ for pure hydrogen and $l_{fl}/D \approx 10$ for natural gas. These results agree well to the flame lengths measured by zur Nedden et al. for a geometrically similar single-jet burner without swirl, when considering the difference in total mass flowrate [9]. Consequently, the characterized fluidically variable-swirl burner is found to suitably realize nonswirled jet flames, namely, at a set MFR value of 9%. This effect is equally perceptible in the flame shapes derived from operating conditions at a lower thermal power output (at $P_{th} = 80$ kW, not shown).

For nonreacting conditions, the zero-swirl MFR value was found at 6%, see Fig. 1 or Ref. [17]. This offset is caused by the higher temperature level of the preheated primary air compared to the secondary air, leading to a reduced flow momentum ratio. Additionally, the injection of fuel gas introduces an additional axial momentum, diminishing the maximum range of inducible swirl. A more detailed discussion of the concept's different behavior between cold and reacting experiments is provided in Sec. 3.3.

3.1.2 Effects of Mass Flow Ratio on the Flame Shapes. Starting from the operating condition at MFR = 9%, as discussed above, the MFR value is varied in the range specified in Table 1. As assessed in the development process of the burner, a higher MFR value corresponds to a stronger effect of the wall-adjacent jets within the burner's swirler segment, consequently resulting in a more pronounced deflection of the primary air flow. This results in a higher degree of swirl, and becomes evident when discussing the flame shapes, as exemplified in Fig. 6. With increasing MFR, for all investigated hydrogen blends, the flame length is significantly reduced. This effect is especially visible within the transition between MFR = 9% and 13%. At MFR = 13%, the flame is subjected to a low level of swirl, significantly reducing its respective length, especially for low X_{H_2} values. The swirl, however, does not invoke a vortex breakdown and thus does not yet result in a stabilized V-shaped flame. In contrast to the condition at MFR = 9% without swirl, the operating point at MFR = 13% yields more compact flames, stabilizing in a spherical shape. With further increasing MFR, the flame is increasingly stabilized and encompasses a pronounced V-shape, typical for swirl-stabilized flames. In this respect, for operating conditions with low hydrogen contents, the flame shape is comparable to results acquired for state-of-the-art swirl-stabilized burners, as, e.g., shown in Ref. [12] by Kuhn et al. for a burner comparable in geometry and operational range, at a swirl number of $S \approx 0.9$. With increasing MFR, consequently increasing swirl, and thus an invigorated recirculation zone, the magnitude of the tip angle of the V-shaped cone is reduced. This effect, however, saturates for MFR values $\geq 25\%$.

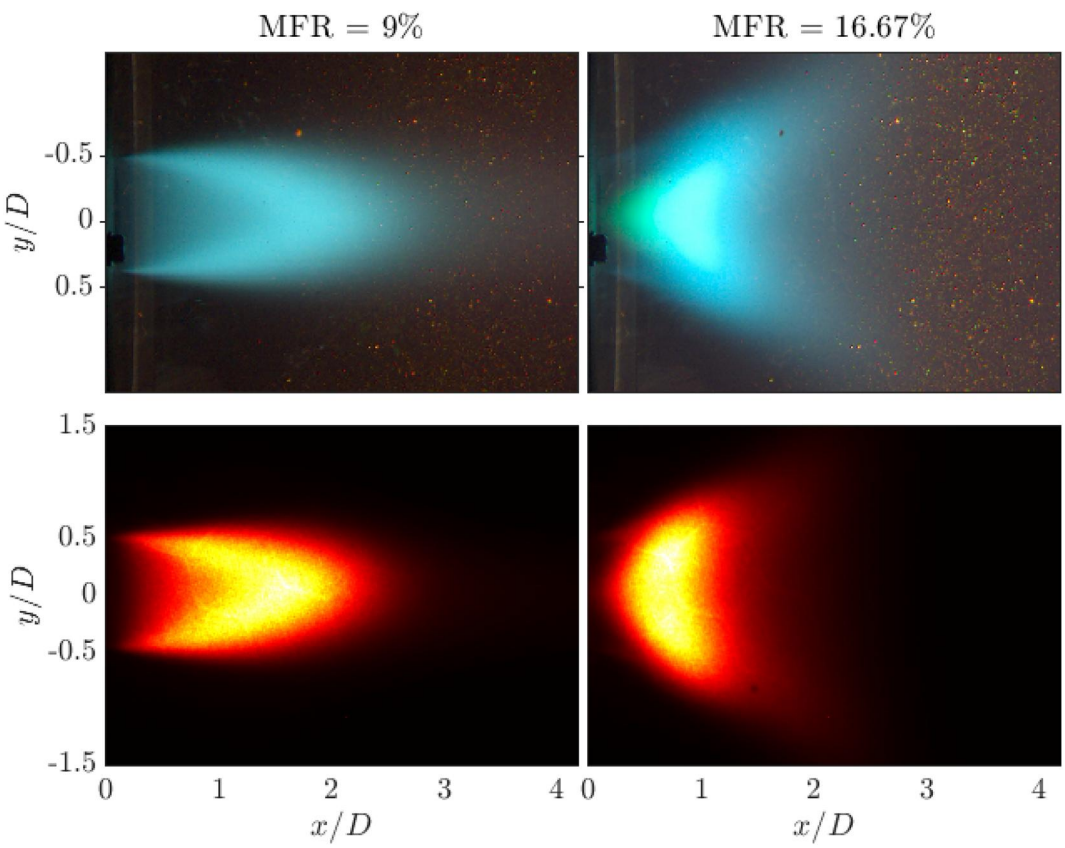

Fig. 5 Exemplary visualization of an operating point at P_{th} = 100 kW and X_{H_2} = 50% for two swirl cases: Axial jet (left) and swirled flow (right). Top row shows images within the visible spectrum, bottom row shows mean OH* images.

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/1/011024/7536570/gtp-25-1257.pdf by guest on 01 January 2026

Similarly, for MFR values below MFR = 9%, the flames undergo a change in shape, with the flame length shortening as MFR decreases. As assessed numerically and experimentally for non-reacting conditions (see Refs. [17,18]), the operating region below MFR_0 exhibits a counter-rotating swirl. The observed flame shapes in Fig. 6 suggest that this behavior persists under reacting conditions, as the flame shape changes in a similar manner with increasing and decreasing MFR around MFR = 9%.

3.1.3 Effects of X_{H_2} on the Flame Shapes. As expected, the flame shapes vary with the hydrogen share. With increasing hydrogen content, the flame shortens and stabilizes closer to the nozzle outlet. This is due to the increased reactivity and higher flame speed associated with hydrogen. This is in accordance with findings of studies investigating the effects of hydrogen content on flame shapes (e.g., see Ref. [21]). From Fig. 6, it becomes apparent that not every operating condition could be sufficiently stabilized to acquire the respective flame shapes. For the investigated operating conditions, the flame is susceptible to flame flash-back events for conditions combining a high degree of swirl and a high hydrogen content. This flash-back dependence limits the relevant operating range of the characterized burner, and is further discussed in Sec. 3.3.

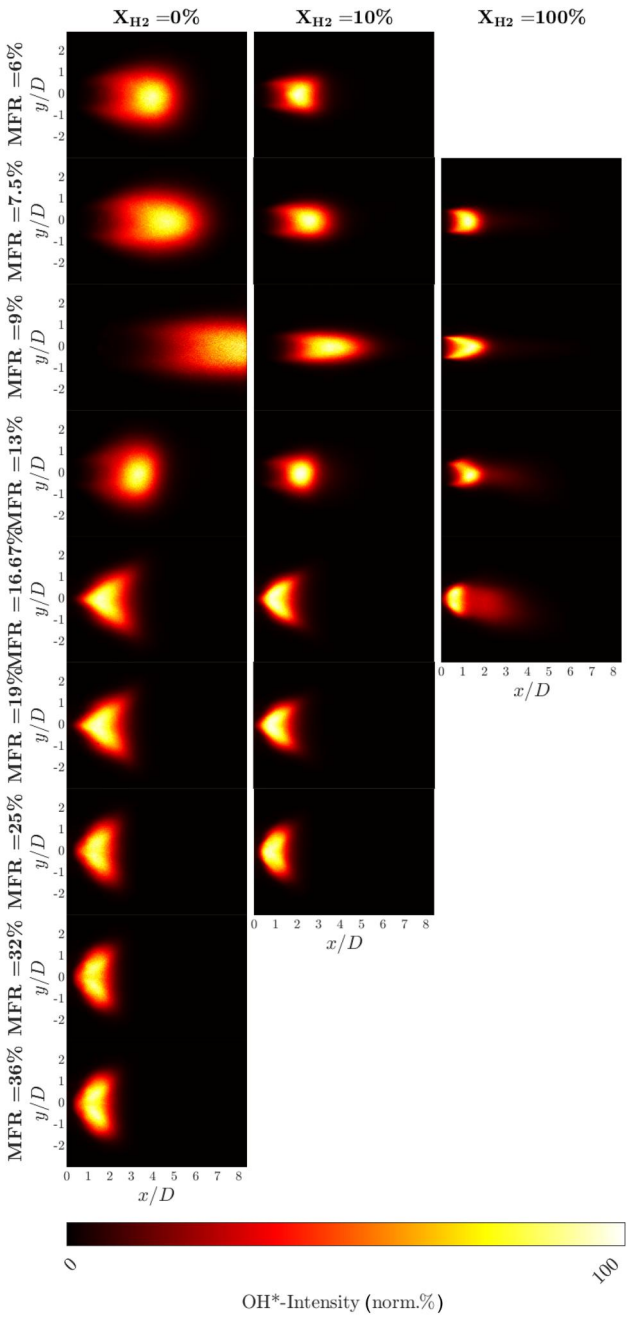

Fig. 6 Flame shapes by means of averaged OH* Images for P_{th} = 100 kW for X_{H_2} = 0%, 10%, and 100% (mass fraction) over all investigated MFR values

3.1.4 Static Flame Stability. To evaluate the static stability of each operating point, the standard deviation of the mean emitted light intensity is calculated for each OH* image pixel, serving as an indicator to quantify and localize regions of flame fluctuations. While the line-of-sight integration in the OH* recording can cause some fluctuations to cancel out, axial variations that are uniform in the azimuthal direction are effectively captured. As variations in MFR influence the swirl number and, consequently, the axial flame extent, see Fig. 6, the computed standard deviation effectively captures the most critical fluctuations. Figure 7 exemplifies the standard deviation of the acquired OH* images for selected operating conditions at $P_{th} = 100$ kW. The blue lines show the location at which the mean heat release is 50% of its respective maximum values, marking the location of the flame.

The largest spatial extent of fluctuations is observed at MFR = 7.5% and low hydrogen content. At MFR values below 9%, the operating point lies within the counterswirl region, previously identified under nonreacting conditions as a region of high unsteadiness, making it impractical for stable operation [17]. The significant spatial extent of OH* fluctuations in this operating range observed in Fig. 7 confirms this behavior.

At the zero-swirl point ($MFR_0 = 9\%$), the flames are of conical shape. Under this operating condition, the spatial extent of fluctuations is high for natural gas but decreases significantly with increasing hydrogen content. This is because, at low hydrogen content, the flame is detached from the burner base plate and stabilizes in the free shear flow. As the hydrogen content increases, the flame anchors at the outlet of the burner nozzle, where it is less susceptible to flow fluctuations. This trend can also be recognized for MFR = 7.5%. At MFR = 16.67%, an increase in hydrogen initially reduces the spatial extent of fluctuations. With pure hydrogen, however, the fluctuations spread out again over a larger area. This behavior occurs because the MFR = 16.67% pure hydrogen case operates near the flashback limit, resulting in increased unsteadiness.

At lower hydrogen contents and thus greater distance from the flashback limit ($X_{H2} = 0\%$ and $X_{H2} = 10\%$), the spatial extent of OH* fluctuations decreases when increasing the MFR value from 9% to 16.67%. The increasing MFR and thus the increasing air rotation reduce the axial flow velocity at the flame's centerline, eventually inducing vortex-breakdown and stabilizing the flame in a V-shape. In this state, the spatial extent of the OH* disturbances is smaller, while the magnitude of the disturbance increases with the compactness of the V-flame. The high magnitude of heat release fluctuation is likely caused by unsteadiness of the reversed flow region in the center of the vortex breakdown region. This indicates that the swirl number is not high enough to create a sustained vortex breakdown and the flow field is subjected to intermittency. At even higher MFR values of $\geq 25\%$, however, the imprinted air rotation is sufficient to stabilize the vortex breakdown, leading to overall low fluctuations in heat release.

3.2 Emissions. As mentioned in Sec. 2.2.1, the raw emission data is corrected to render it quantitatively comparable between different fuel compositions. In this respect, the correction routine introduced by Garan et al. [22] is followed, which generalizes and expands the standard volumetric correction to a reference exhaust oxygen level, originally based on natural gas/methane fuels. This correction allows for the comparison of emission levels acquired for a multitude of differing combustion processes, including air or inert gas dilution, oxygen enrichment, fuel variations, and different thermal powers. The correction routine is exemplary shown for NO_x

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/1/011024/7536570/gtp-25-1257.pdf by guest on 01 January 2026

emissions in the following. First, the raw dry concentrations, $NO_{x,raw}$, are corrected to a customary reference oxygen level of $O_{2,ref} = 15\%$:

$$NO_{x,ref-O_2} = NO_{x,raw} \cdot \left(\frac{0.21 - O_{2,ref}}{0.21 - O_{2,meas.}}\right) \tag{2}$$

with $O_{2,meas.}$ denoting the measured oxygen level within the exhaust gas. The oxygen-corrected emissions $NO_{x,ref-O_2}$ are further corrected, accounting for the specific fuel gas composition in combination with the definition of a reference fuel gas (denoted with the subscript "ref"), chosen to be pure methane (CH_4) in the present study. Assuming ideal gas behavior, the corrected emission levels $NO_{x,corr.}$ can be expressed as

$$NO_x = NO_{x,corr.} = NO_{x,ref-O_2} \cdot \left(\frac{\alpha + 1 - 0.21}{\alpha_{ref} + 1 - 0.21}\right) \cdot \left(\frac{\beta'}{\beta'_{ref}}\right) \tag{3}$$

with α and β being fuel-specific constants (see superscript "f"), as defined in the following equation:

$$\alpha = \frac{n_C^f + n_S^f + n_N^f}{\beta} \text{ and } \beta = \frac{n_H^f - n_{Cl}^f}{4} - 2 \cdot \left(n_O^f + n_N^f\right) \tag{4}$$

Here, n_i^f stands for the amount of element i within one fuel molecule. For the reference CH_4 fuel reacting with air, the constants are found to be $\alpha_{ref} = \beta_{ref} = 1$. Additionally, the resulting emission levels are translated into input-based quantities, based on the power output of the used fuel blend. This is accomplished by the introduction of the lower heating value of the specific fuel $\beta'_i = \beta_i/LHV_i$. As also reported by Douglas et al. [23], the normalization by the thermal energy input of the utilized fuels compensates for the overestimation of emissions of high-hydrogen fuels, potentially occurring during the correction of dry emissions to a reference oxygen level.

3.2.1 Effects of X_{H_2} on NO_x Emissions. In order to assess the effect of a swirl variation on the combustion of different fuel blends, ranging from pure H_2 to NG, emission levels are analyzed over a broad operating range. A special focus lies on the emission of NO_x, quantified and discussed in this section, as well as compared to a reference burner. Figure 8 depicts the levels of emitted NO_x over the hydrogen content X_{H_2} and MFR range (i.e., swirl magnitudes) for the investigated operating range. The overall NO_x emission levels are found to be generally lower for the cases featuring a lower thermal power output. For both investigated thermal power levels, however, a similar trend can be observed. The emitted NO_x levels generally increase with increasing hydrogen content within the combusted fuel blend. In the present study, both the thermal power output and

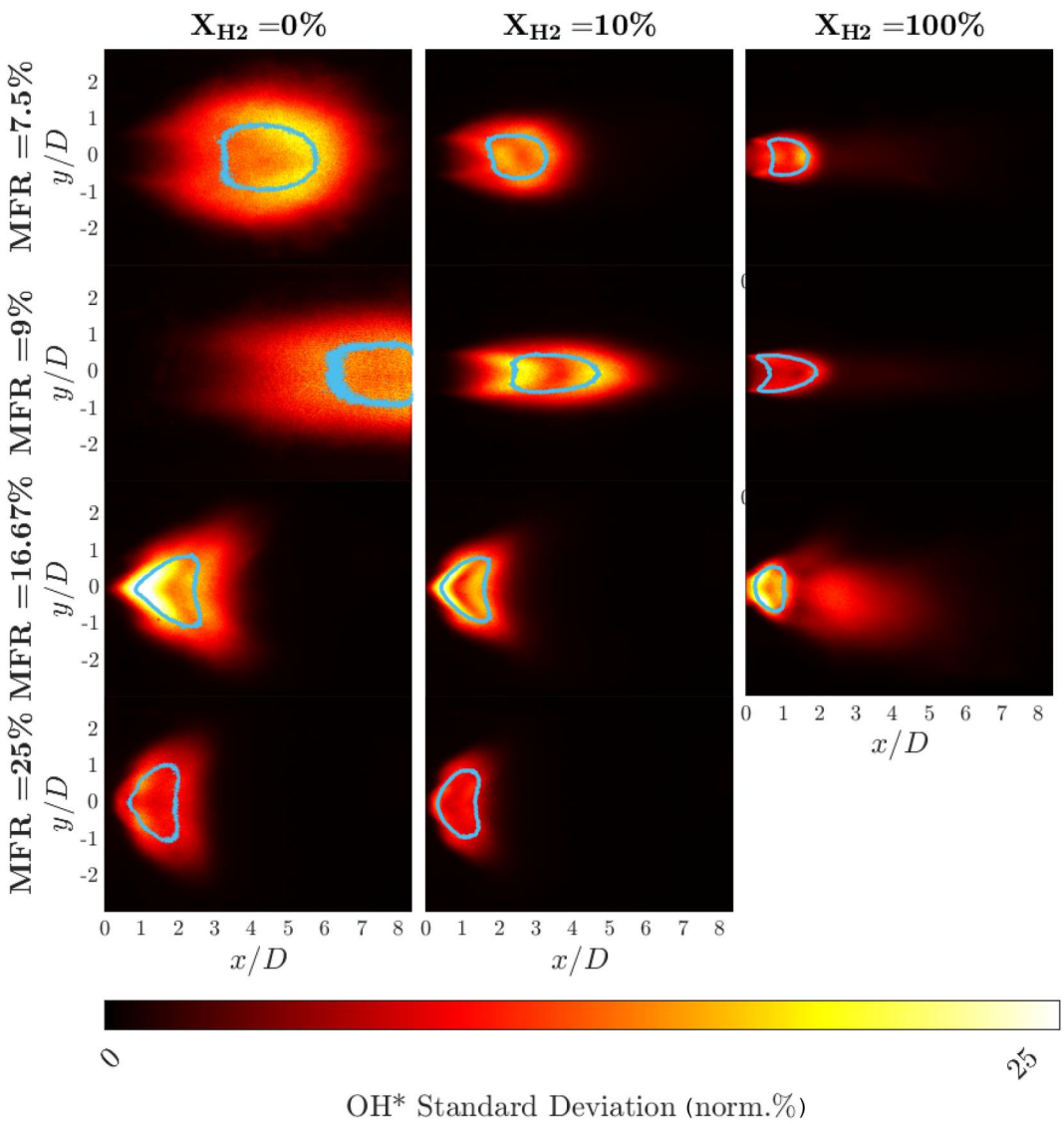

Fig. 7 Deviation of OH* images from the respective mean for selected operating conditions for $P_{th} = 100$ kW. Contours of mean flame shapes (50% of max. intensity) are delineated.

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/1/011024/7536570/gtp-25-1257.pdf by guest on 01 January 2026

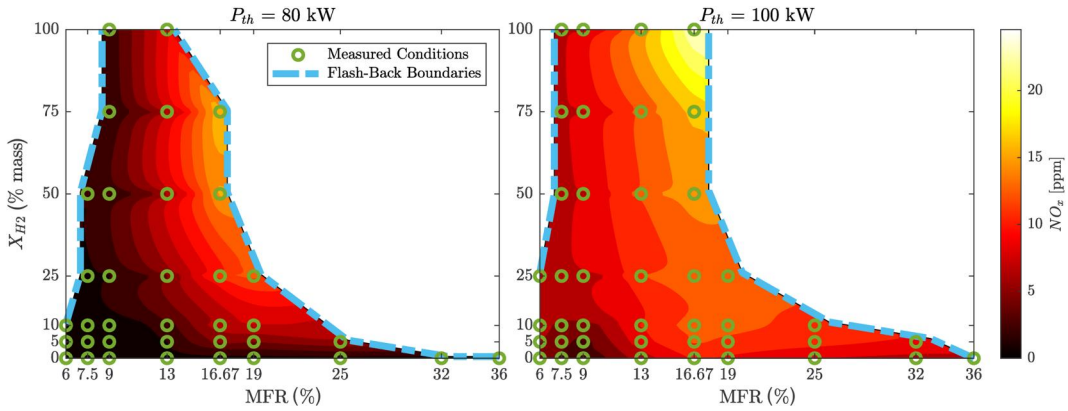

Fig. 8 Corrected NO_x values over investigated operating range. Interpolation based on acquired measurement conditions: For P_{th} = 80 kW (left) and P_{th} = 100 kW (right).

the total amount of combustion air is kept constant. Consequently, the equivalence ratio decreases with increasing hydrogen content, as shown in Fig. 4. As a result, the respective adiabatic flame temperature is approximately kept at a constant level (at approx. 1540 K for $P_{th} = 80$ kW and 1770 K for $P_{th} = 100$ kW), thus not accounting for the increase of NO_x emissions with increasing X_{H_2}.

With increasing X_{H_2}, however, the respective flames become increasingly compact. The spatial temperature distribution in the reaction zone can be assumed to confine accordingly, analogously leading to more pronounced local temperature maxima. With other combustion parameters remaining approximately unaltered, this effect is likely to account for the steady increase of NO_x emissions with increasing X_{H_2}.

The overall NO_x emissions are comparable to emission levels acquired for state-of-the-art burners, not featuring the possibility of swirl variation. In Ref. [9], zur Nedden et al. investigated the effect of the hydrogen content on NO_x emissions of a single-jet-burner, not featuring any swirl. The jet burner was operated at conditions comparable to those considered in the present study and yielded NO_x emission levels between 4.5 ppm and 6.8 ppm, which is approximately 30% lower than the levels measured in the present work. However, the jet burner features a mixing tube with a length of approximately ten nozzle diameters, allowing for a higher degree of fuel-air mixing, accounting for the observed offset. Furthermore, the NO_x emission levels of the novel fluidically variable swirl burner are found to comply well with the results found in literature for swirl-stabilized burners. In Ref. [8], Göke and Paschereit studied the influence of steam dilution on NO_x emissions in a swirl-stabilized burner for different hydrogen–methane fuel blends. Again, the NO_x emission levels are comparable to the present study, especially when comparing cases with a higher MFR, i.e., with elevated swirl. Consequently, over its entire envisaged operating range, the burner is found to exhibit comparable emission characteristics to state-of-the-art swirl-stabilized and jet burners.

What becomes evident, is that the level of emitted NO_x is strongly linked to the MFR value, i.e., swirl. This effect is maximized toward higher hydrogen concentrations. For both thermal power outputs, at the same X_{H_2}, the amount of emitted NO_x can differ up to

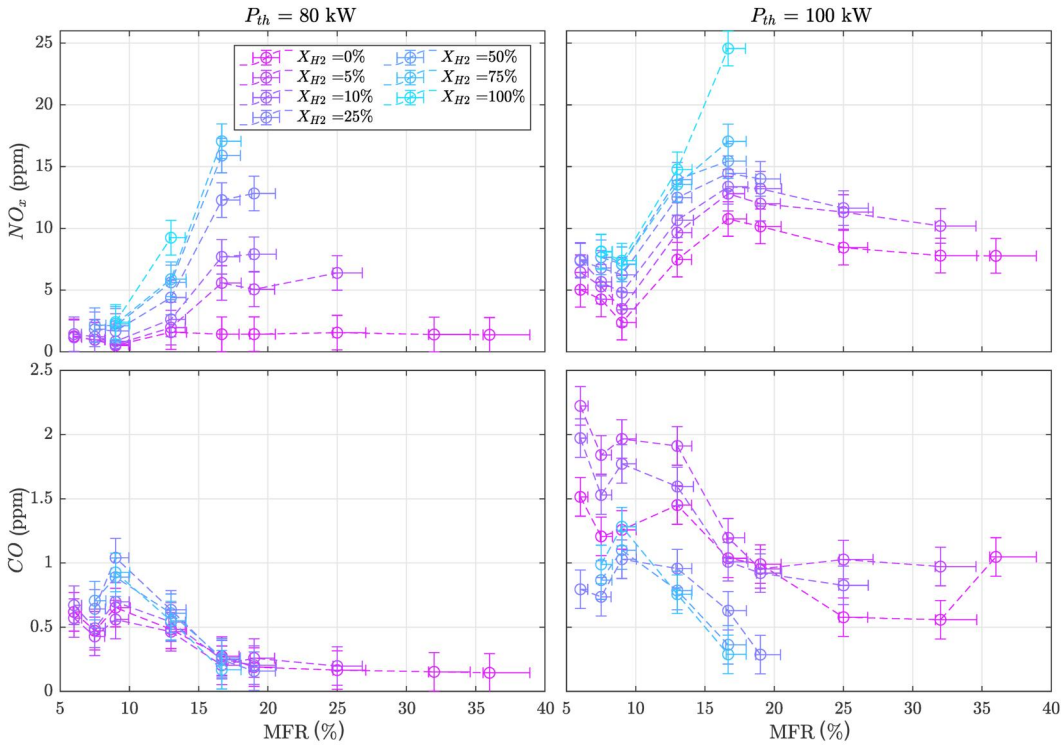

Fig. 9 Corrected NO_x (top row) and CO (bottom row) emission levels over the varied MFR: For P_{th} = 80 kW (left column) and P_{th} = 100 kW (right column)

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/1/011024/7536570/gtp-25-1257.pdf by guest on 01 January 2026

approximately 15 ppm and 17 ppm, respectively, between conditions with a low and a high degree of swirl, amounting to a relative difference of over 60% of the acquired value range. Since the degree of swirl is observed to have a significant effect, the dependency of the MFR on NO_x emissions is further discussed in the Sec. 3.2.2.

3.2.2 Effects of Mass Flow Ratio on NO_x Emissions. The effect that the MFR, and thus the magnitude of swirl, has on the emitted NO_x is shown in the top row of Fig. 9. Here, the experimental uncertainty in the acquisition of emission data is depicted as error bars. It represents the error introduced by the employed data acquisition and processing chain, including emission analysis and flow conditioning hardware. For all hydrogen mass fractions X_{H_2}, the operating condition at $MFR_0 = 9\%$, denoting the nonswirled axial jet flame, stands out as the operational minimum of emitted NO_x. The axial jet involves the highest axial velocity magnitudes at the flame's centerline and, thus, reduces the residence time within the combustion region, especially when compared to conditions involving swirled flows. Based on the experimentally deduced flame length (see Fig. 7) and the jet velocity, residence times for the nonswirled (MFR = 9%) flame with $P_{th} = 100$ kW and $X_{H_2} = 10\%$, combustion residence times amount to approx. 2 ms. The X_{H_2} dependency of NO_x emissions is minimized at MFR_0 and increases with the magnitude of swirl, including the unsteady conditions with counter-rotating swirl for MFR < 9%. With increasing MFR, NO_x emissions are subjected to a significant increase, linked to the corresponding increase in swirl magnitude. Here, the reduced axial velocity augments the residence time within the flame, which, at the same time, is increasingly compacted. For comparison, reaction residence times for a geometrically similar burner with $S \approx 0.7$, $P_{th} = 81$ kW, $\phi = 0.56$, and a total mass flow of 180 kg/h are identified in a range of 25 ms–35 ms [24,25]. This increase in NO_x emissions, however, is mitigated after reaching an operational maximum. Here, due to the elevated degree of swirl, the induced vortex-breakdown and the respective recirculation zones maximize the reaction residence time. For further increasing MFR, NO_x emissions decrease, which can be explained by the improving quality of air–fuel mixing. This effect appears to further decrease the emitted NO_x for even further increased MFR values (MFR > 20%), until reaching MFR conditions at which the effect on NO_x becomes negligible. Given the operational limits in terms of MFR, this region is only resolved for low hydrogen contents ($X_{H_2} \leq 5\%$). Here, however, the findings comply well with the acquired OH* flame shapes which, comparably, are not subjected to further changes for MFR ≥ 25%.

3.2.3 Effects of Mass Flow Ratio on CO Emissions. The bottom row within Fig. 9 exemplifies the corrected CO emission levels for $P_{th} = 80$ and 100 kW. Within the experimental uncertainty, the emitted CO is negligible for $X_{H_2} = 100\%$, as the combusted fuel does not include any fraction of carbon, and is therefore not shown. With decreasing X_{H_2}, the content of elemental carbon increases with increasing share of NG. Accordingly, the level of emitted CO increases. This increase, however, is found to be limited within the experimental uncertainty, as the overall levels of emitted CO are low. Especially for conditions at $P_{th} = 80$ kW, with a comparably lower absolute share of carbon within the combustible fuel, the maximum CO levels amount to approx. one ppm, hampering the distinction of quantitative trends. The MFR value, on the other hand, appears to have a more pronounced influence on the emitted CO. Contrary to the emitted NO_x, the acquired CO levels generally decrease with increasing MFR. At MFR ≤ 9%, i.e., in the low-swirl regime, the highest CO levels are acquired. The comparably low residence times are not sufficient for a complete carbon conversion, especially with regard to the prevailing flame temperatures. With further increasing MFR, and accordingly increasing residence times, the emission levels steadily decrease for all investigated fuel blends. Analogously to the trend observed for NO_x emissions, for MFR ≥ 20%, the CO decrease as well as the effect of MFR are mitigated.

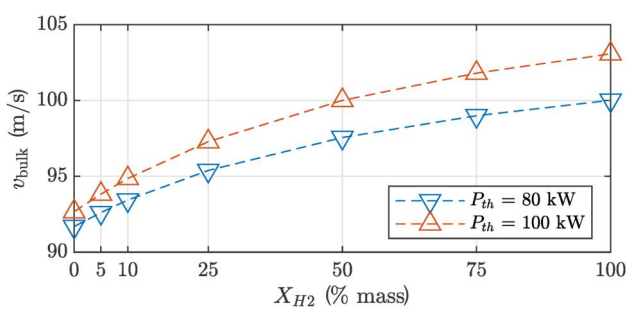

Fig. 10 Bulk-flow velocity over X_{H_2}

3.3 Operational Range. From the experimental results, a technically relevant operating range can be derived. In addition to the measured levels of NO_x emissions, Fig. 8 includes the boundaries at which the operation is limited by flame flashback (blue dash-dotted lines). Notably, the resultant operational envelope is narrower for the operation at $P_{th} = 80$ kW compared to the operation at 100 kW. More conditions with $X_{H_2} \geq 10\%$ resulted in flashback events for MFR ≠ MFR_0, despite the comparably lower equivalence ratios. This effect originates from the reduced amount of fuel being axially injected at the base of the burner's swirler segment for lower thermal power levels. This results in a lower bulk-flow velocity, and thus in a reduced axial centerline velocity. Figure 10 shows the calculated bulk-flow velocity v_{bulk} over the investigated X_{H_2} range.

Especially for high-hydrogen contents, v_{bulk} is considerably higher for the 100 kW case. For $P_{th} = 80$ kW, the axial centerline velocity is accordingly reduced, increasing the risk of vortex-induced flash-back.

At conditions with $MFR_0 = 9\%$, a nonswirled jet-stabilized flame can be observed (see Figs. 6 and 7), featuring a low level of spatial heat release fluctuation at high hydrogen contents and denoting an operational minimum of the measured NO_x emissions. Here, for high levels of X_{H_2}, compact and stable flames can be observed. For flames combusting a higher share of NG ($X_{H_2} < 10\%$), however, the flame length and combustion region are increased, causing the flame to be affected by flow disturbances, which can cause slight offsets to the axial swirl distribution. On the other hand, for conditions with MFR ≥ 25%, fully swirl-stabilized flames are observed, characterized by significant compactness and manifesting an expected V-shape. Here, especially for low hydrogen contents and higher MFR conditions, the heat release fluctuations are significantly reduced, indicating a stable combustion. For high-hydrogen cases ($X_{H_2} \geq 50\%$), however, increased NO_x emissions are observed as well as a higher flashback probability.

From the cold characterization (see Fig. 1), an apparent difference lies in the offset of MFR_0 with $MFR_{0,cold} = 6\% < 9\% = MFR_{0,hot}$. Here, the preheated primary air results in a reduced flow momentum ratio between secondary and primary air flow $(\dot{J}_s/\dot{J}_p)_{hot} \approx 44\% \cdot (\dot{J}_s/\dot{J}_p)_{cold}$. For the same level of flow control, mathematically, an enhanced MFR would be needed, amounting to $MFR_{0,hot} = 1/44\% \cdot MFR_{0,cold} \approx 13.7\%$. However, with $MFR_{0,hot} = 9\%$, the experimentally observed $MFR_{0,hot}$ is significantly lower. This offset is likely to be caused by the additional flow momentum of the axial fuel injection. For higher MFR, the additional axial momentum counteracts the swirl-inducing efficiency. However, due to the high injection velocity of the secondary air mass flow m_s (determining the MFR), the maximum axial fuel momentum amounts to approx. 20% of $\dot{J}_s$ for MFR = 9% and to approx. 2% for MFR = 32%, limiting its effect. As a result, compared to the cold characterization, an MFR offset of $\Delta MFR \approx (13.7\% - 6\%) = 7.7\%$ can be assumed. An accurate quantification of swirl numbers and their relation to MFR, however, requires the investigation of the reacting flow fields by means of suitable measurement techniques.

Consequently, for both thermal powers, conditions featuring a high degree of swirl are identified, advantageously stabilizing flames at low hydrogen levels ($X_{H_2} \leq 10\%$ at MFR ≥ 25%). For

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/1/011024/7536570/gtp-25-1257.pdf by guest on 01 January 2026

hydrogen contents $X_{H_2} \geq 10\%$, the operation at $MFR_0 = 9\%$ is found to be best suitable in terms of emissions and flame stability.

4 Conclusions

In this study, the experimental characterization of a fluidically variable-swirl-stabilized burner was presented. The working principle relies on fluidic thrust vectoring methods, studied in the field of aircraft propulsion systems. The concept's capability has been experimentally assessed in reacting conditions. The experimental investigation consisted of the acquisition of OH* images and exhaust emission data. The investigated operating range spanned thermal power outputs of 80 kW and 100 kW, NG-hydrogen fuel blends with hydrogen contents between 0% and 100% (by mass), and MFR between 6% and 36%.

The behavior and performance of the novel burner have been assessed and discussed. The observed span of characteristics resembles both the behavior of state-of-the-art swirl-stabilized and jet-stabilized burners in terms of flame shapes, emissions, and operational stability. Additionally, an operating range was experimentally identified, allowing for a technically relevant operation of the characterized burner in terms of achievable swirl, its active control, and the applicability to a varying hydrogen content within the combusted fuel. Within the identified range, the investigated burner is capable of generating and stabilizing nonswirled jet flames, as well as fully swirl-stabilized flames. This variation of flow and combustion regime is found to affect the NO_x and CO emissions. For all operating conditions, however, an operational optimum of NO_x emissions could be identified, consisting of a suitable MFR, and hence of a specific degree of swirl. Based on the MFR, the swirl variation can be performed dynamically. Advantageously, the quality of the combustion process is not negatively affected by the MFR, thus, by the degree of swirl, as shown from the acquired low CO concentrations.

In summary, the experimental characterization of the presented burner demonstrated its swirl-varying capability within technically relevant limits. The concept has the potential to facilitate fuel flexibility of combustors within power generation systems as well as in academic research. Due to its stable operation, the possibility of dynamical swirl variation, and its resulting fuel flexibility, the concept advantageously offers multifaceted possibilities to further assess the effect of swirl on combustion efficiency, emissions, as well as on flame dynamics and stability.

Acknowledgment

The authors are grateful to Christoph Peisdersky, Katharina Schmidt, David Lück, Andy Göhrs, and Thorsten Dessin for their support in the completion of this work. Special acknowledgment is given to Markus Malcher and Professor Dr.-Ing. Dirk Oberschmidt, Chair of Micro-and Precision Devices at Technische Universität Berlin, for their outstanding expertise in manufacturing key burner components.

Funding Data

- European Research Council (ERC) as part of the Advanced Grant HYPOTHESis (Grant ID: 101019937; Funder ID: 10.13039/501100000781).

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

D = diameter of mixing tube/swirler outlet (m)
$\dot{J}$ = flow momentum rate (N)
l = length (m)
$\dot{m}$ = mass flow (kg s^{-1}, kg h^{-1})
MFR = mass flow ratio (secondary to primary) (%)
n = molar amount within one molecule
P_{th} = thermal power (kW)
S = swirl number
v = velocity (m s^{-1})
X = mass fraction (content) (%)
x, y, z = Cartesian coordinate positions (m)
α, β, β' = chem. educt/product-specific constants
Δ = difference
λ = wavelength (nm)
ϕ = equivalence ratio

Abbreviations

ERC = European Research Council
LHV = lower heating value
NG = natural gas
UV = ultraviolet

Superscripts and Subscripts

bulk = bulk flow
cold = non-reacting
corr = corrected
f = fuel
fl = flame
hot = reacting
meas = measured
p = primary flow
s = secondary flow
raw = raw and dry
ref = reference
tot = total
0 = non-swirled/axial

Chemical Terms

C = elementary carbon
CH_4 = methane; concentration (% mol.)
Cl = elementary chlorine
CO = carbon monoxide; concentration (ppm)
CO_2 = carbon dioxide; concentration (% mol.)
H = elementary hydrogen
N = elementary nitrogen
NO_x, NO_2, NO = nitrogen oxides; concentration (ppm)
O = elementary oxygen
OH* = hydroxide radical; intensity (norm. %)
O_2 = oxygen; concentration (% mol.)
S = elementary sulfur

References

[1] Syred, N., and Beér, J. M., 1974, "Combustion in Swirling Flows: A Review," Combust. Flame, **23**(2), pp. 143–201.
[2] Mattingly, J., Von Ohain, H., and Schetz, J., 2006, *Elements of Propulsion: Gas Turbines and Rockets*, 2nd ed., American Institute of Aeronautics and Astronautics, Inc., Reston, VA.
[3] Durox, D., Moeck, J., Bourgouin, J.-F., Morenton, P., Viallon, M., Schuller, T., and Candel, S., 2013, "Flame Dynamics of a Variable Swirl Number System and Instability Control," Combust. Flame, **160**(9), pp. 1729–1742.
[4] Chigier, N., and Chervinsky, A., 1967, "Experimental Investigation of Swirling Vortex Motion in Jets," ASME J. Appl. Mech., **34**(2), pp. 443–451.
[5] Lilley, D., 1977, "Swirl Flows in Combustion: A Review," AIAA J., **15**(8), pp. 1063–1078.
[6] Feikema, D., Chen, R.-H., and Driscoll, J., 1990, "Enhancement of Flame Blowout Limits by the Use of Swirl," Combust. Flame, **80**(2), pp. 183–195.
[7] Stöhr, M., Boxx, I., Carter, C., and Meier, W., 2011, "Dynamics of Lean Blowout of a Swirl-Stabilized Flame in a Gas Turbine Model Combustor," Proc. Combust. Inst., **33**(2), pp. 2953–2960.
[8] Göke, S., and Paschereit, C., 2013, "Influence of Steam Dilution on Nitrogen Oxide Formation in Premixed Methane/Hydrogen Flames," J. Propul. Power, **29**(1), pp. 249–260.
[9] Zur Nedden, P., Eck, M., Lückoff, F., Panek, L., Orchini, A., and Paschereit, C., 2023, "Flame Transfer Function and Emissions of a Piloted Single Jet Burner: Influence of Hydrogen Content," ASME Paper No. GT2023-103032.

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/1/011024/7536570/gtp-25-1257.pdf by guest on 01 January 2026

[10] Sheen, H., Chen, W., Jeng, S., and Huang, T., 1996, "Correlation of Swirl Number for a Radial-Type Swirl Generator," Exp. Therm. Fluid Sci., **12**(4), pp. 444–451.
[11] Gupta, A. K., Ramavajjala, M. S., Chomiak, J., and Marchionna, N., 1991, "Burner Geometry Effects on Combustion and NOx Emission Characteristics Using a Variable Geometry Swirl Combustor," J. Propul. Power, **7**(4), pp. 473–480.
[12] Kuhn, P., Terhaar, S., Reichel, T., and Paschereit, C., 2015, "Design and Assessment of a Fuel-Flexible Low Emission Combustor for Dry and Steam-Diluted Conditions," ASME Paper No. GT2015-43375.
[13] Deere, K., 2003, "Summary of Fluidic Thrust Vectoring Research at NASA Langley Research Center," AIAA Paper No. 2003-3800.
[14] Mason, M., and Crowther, W., 2002, "Fluidic Thrust Vectoring for Low Observable Aircraft," CEAS Aerospace Aerodynamic Research Conference, Cambridge, UK, June 10–12, pp. 10–12.
[15] Mason, M., and Crowther, W., 2004, "Fluidic Thrust Vectoring for Low Observable Air Vehicles," AIAA Paper No. 2004-2210.
[16] Banazadeh, A., Saghafi, F., Ghoreyshi, M., and Pilidis, P., 2008, "Experimental and Computational Investigation Into the Use of co-Flow Fluidic Thrust Vectoring on a Small Gas Turbine," Aeronaut. J., **112**(1127), pp. 17–25.
[17] Eck, M., Zur Nedden, P., von Saldern, J., Orchini, A., and Paschereit, C., 2023, "Design of a Swirl-Stabilized Burner With Fluidically Variable Swirl Number," ASME Paper No. GT2023-103062.
[18] Eck, M., Zur Nedden, P., von Saldern, J., Peisdersky, C., Orchini, A., and Paschereit, C., 2025, "Experimental Design Validation of a Swirl-Stabilized Burner With Fluidically Variable Swirl Number," ASME J. Eng. Gas Turbines Power, **147**(4), p. 041017.
[19] Guyot, D., Guethe, F., Schuermans, B., Lacarelle, A., and Paschereit, C., 2010, "CH*/OH* Chemiluminescence Response of an Atmospheric Premixed Flame Under Varying Operating Conditions," ASME Paper No. GT2010-23135.
[20] Guethe, F., Guyot, D., Singla, G., Noiray, N., and Schuermans, B., 2012, "Chemiluminescence as Diagnostic Tool in the Development of Gas Turbines," Appl. Phys. B, **107**(3), pp. 619–636.
[21] Beuth, J., Reumschüssel, J., von Saldern, J., Cosić, B., Paschereit, C., and Oberleithner, K., 2025, "Experimental Study of the Dynamics of Lean Premixed Hydrogen Flames in a Multi Jet Combustor," ASME J. Eng. Gas Turbines Power, **147**(3), p. 031013.
[22] Garan, N., Dybe, S., Paschereit, C., and Djordjevic, N., 2023, "Consistent Emission Correction Factors Applicable to Novel Energy Carriers and Conversion Concepts," Fuel, **341**, p. 127658.
[23] Douglas, C., Shaw, S., Martz, T., Steele, R., Noble, D., Emerson, B., and Lieuwen, T., 2022, "Pollutant Emissions Reporting and Performance Considerations for Hydrogen–Hydrocarbon Fuels in Gas Turbines," ASME J. Eng. Gas Turbines Power, **144**(9), p. 091003.
[24] Göckeler, K., Terhaar, S., and Paschereit, C., 2014, "Residence Time Distribution in a Swirling Flow at Nonreacting, Reacting, and Steam-Diluted Conditions," ASME J. Eng. Gas Turbines Power, **136**(4), p. 041505.
[25] Göke, S., Schimek, S., Terhaar, S., Reichel, T., Göckeler, K., Krüger, O., Fleck, J., Griebel, P., and Paschereit, C., 2014, "Influence of Pressure and Steam Dilution on NOx and CO Emissions in a Premixed Natural Gas Flame," ASME J. Eng. Gas Turbines Power, **136**(9), p. 091508.

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/1/011024/7536570/gtp-25-1257.pdf by guest on 01 January 2026